
Software evolution

Objectives

- To discuss an approach to assessing evolution strategies for legacy systems

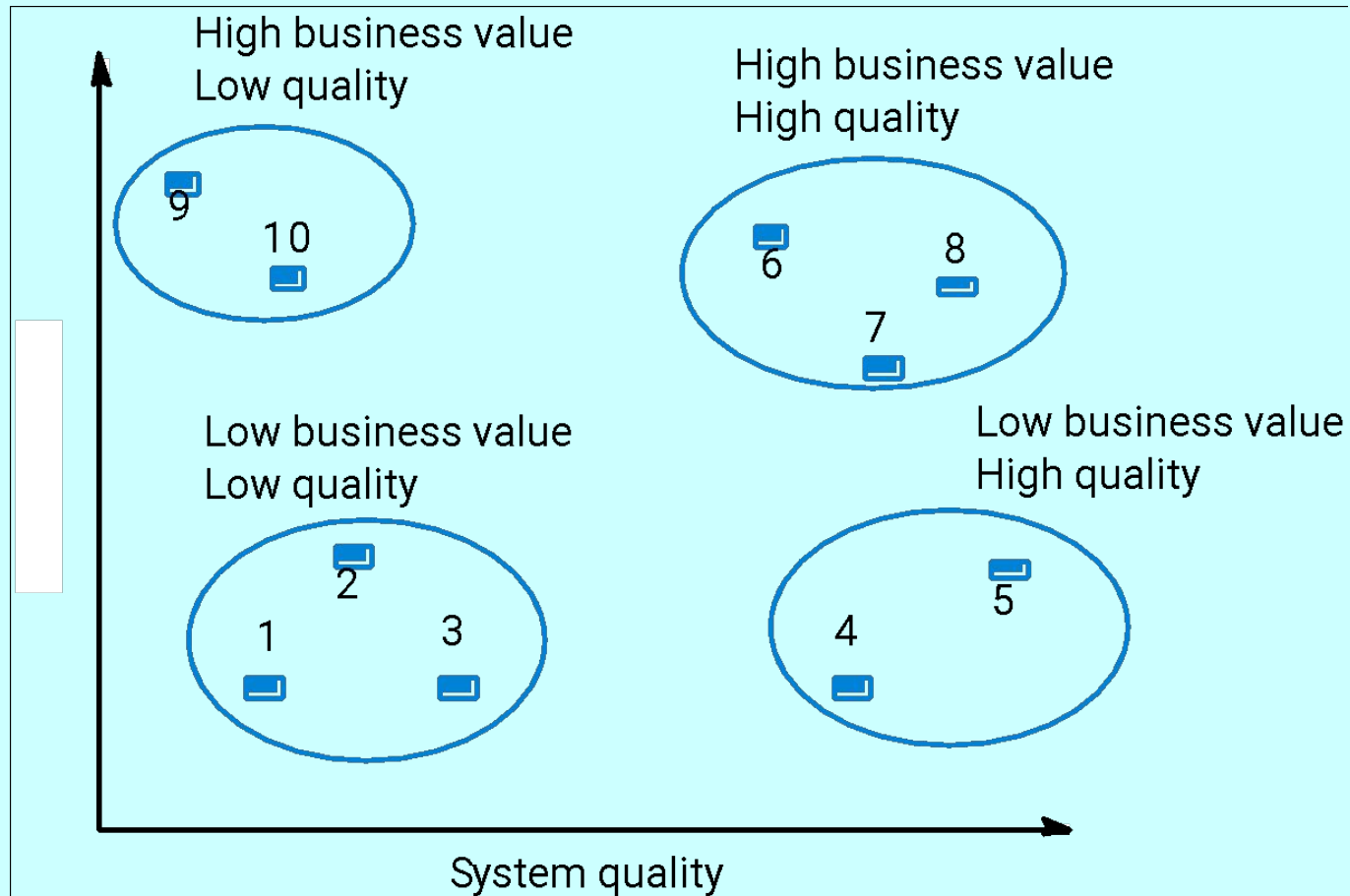
Topics covered

- Legacy system evolution

Legacy system evolution

- Organisations that rely on legacy systems must choose a strategy for evolving these systems
 - Scrap the system completely and modify business processes so that it is no longer required;
 - Continue maintaining the system;
 - Transform the system by re-engineering to improve its maintainability;
 - Replace the system with a new system.
- The strategy chosen should depend on the system quality and its business value.

System quality and business value



Legacy system categories

- Low quality, low business value
 - These systems should be scrapped.
- Low-quality, high-business value
 - These make an important business contribution but are expensive to maintain. Should be re-engineered or replaced if a suitable system is available.
- High-quality, low-business value
 - Replace with COTS, scrap completely or maintain.
- High-quality, high business value
 - Continue in operation using normal system maintenance.

Business value assessment

- Assessment should take different viewpoints into account
 - System end-users;
 - Business customers;
 - Line managers;
 - IT managers;
 - Senior managers.
- Interview different stakeholders and collate results.

System quality assessment

- Business process assessment
 - How well does the business process support the current goals of the business?
- Environment assessment
 - How effective is the system's environment and how expensive is it to maintain?
- Application assessment
 - What is the quality of the application software system?

Business process assessment

- Use a viewpoint-oriented approach and seek answers from system stakeholders
 - Is there a defined process model and is it followed?
 - Do different parts of the organisation use different processes for the same function?
 - How has the process been adapted?
 - What are the relationships with other business processes and are these necessary?
 - Is the process effectively supported by the legacy application software?
- Example - a travel ordering system may have a low business value because of the widespread use of web-based ordering.

Environment assessment 1

Factor	Questions
Supplier stability	Is the supplier is still in existence? Is the supplier financially stable and likely to continue in existence? If the supplier is no longer in business, does someone else maintain the systems?
Failure rate	Does the hardware have a high rate of reported failures? Does the support software crash and force system restarts?
Age	How old is the hardware and software? The older the hardware and support software, the more obsolete it will be. It may still function correctly but there could be significant economic and business benefits to moving to more modern systems.
Performance	Is the performance of the system adequate? Do performance problems have a significant effect on system users?

Environment assessment 2

Support requirements	What local support is required by the hardware and software? If there are high costs associated with this support, it may be worth considering system replacement.
Maintenance costs	What are the costs of hardware maintenance and support software licences? Older hardware may have higher maintenance costs than modern systems. Support software may have high annual licensing costs.
Interoperability	Are there problems interfacing the system to other systems? Can compilers etc. be used with current versions of the operating system? Is hardware emulation required?

Application assessment 1

Factor	Questions
Understandability	How difficult is it to understand the source code of the current system? How complex are the control structures that are used? Do variables have meaningful names that reflect their function?
Documentation	What system documentation is available? Is the documentation complete, consistent and up-to-date?
Data	Is there an explicit data model for the system? To what extent is data duplicated in different files? Is the data used by the system up-to-date and consistent?
Performance	Is the performance of the application adequate? Do performance problems have a significant effect on system users?

Application assessment 2

Programming language	Are modern compilers available for the programming language used to develop the system? Is the programming language still used for new system development?
Configuration management	Are all versions of all parts of the system managed by a configuration management system? Is there an explicit description of the versions of components that are used in the current system?
Test data	Does test data for the system exist? Is there a record of regression tests carried out when new features have been added to the system?
Personnel skills	Are there people available who have the skills to maintain the application? Are there only a limited number of people who understand the system?

System measurement

- You may collect quantitative data to make an assessment of the quality of the application system
 - The number of system change requests;
 - The number of different user interfaces used by the system;
 - The volume of data used by the system.

Key points

- The business value of a legacy system and its quality should determine the evolution strategy that is used.